UNIT-I
Importance of Information System
CAP404
INFORMATION SECURITY AND PRIVACY

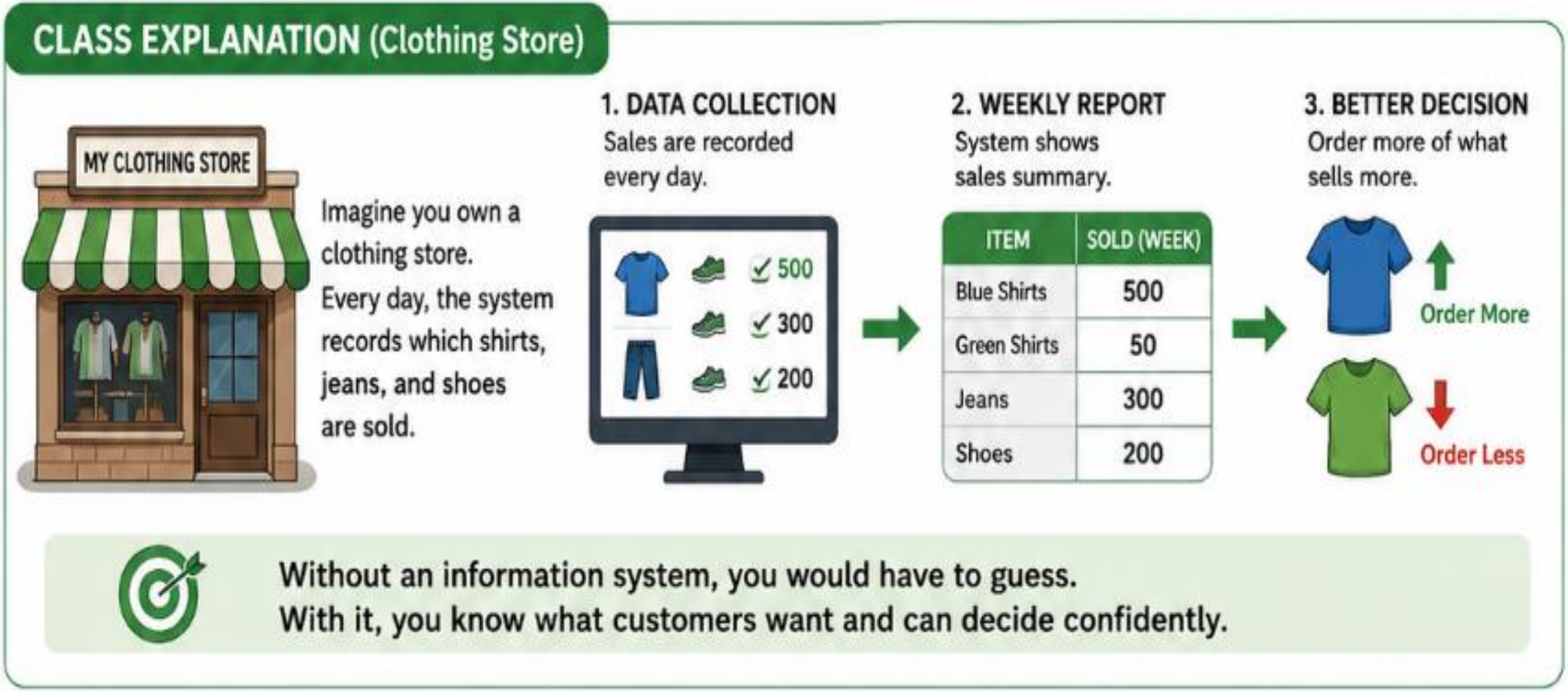
SCHOOL OF COMPUTER APPLICATIONS

DOMAIN: AI &ML

Importance of Information Systems

- **Improves decision-making** by providing accurate and timely information.
- **Increases efficiency through automation** of routine tasks and processes.
- **Enhances communication** within organizations and with customers or partners.
- **Supports business operations** such as inventory, sales, finance, and employee management.
- **Provides a competitive advantage** by improving productivity, innovation, and customer service.

1. Improves Decision-Making



REAL-WORLD EXAMPLE (Amazon)

Amazon tracks millions of customer purchases every day.

1. CUSTOMER PURCHASE DATA

Many people in a city start buying umbrellas.

2. INFORMATION SYSTEM ANALYSIS

The system identifies the trend and predicts higher demand for umbrellas in that city.

3. SMART DECISION

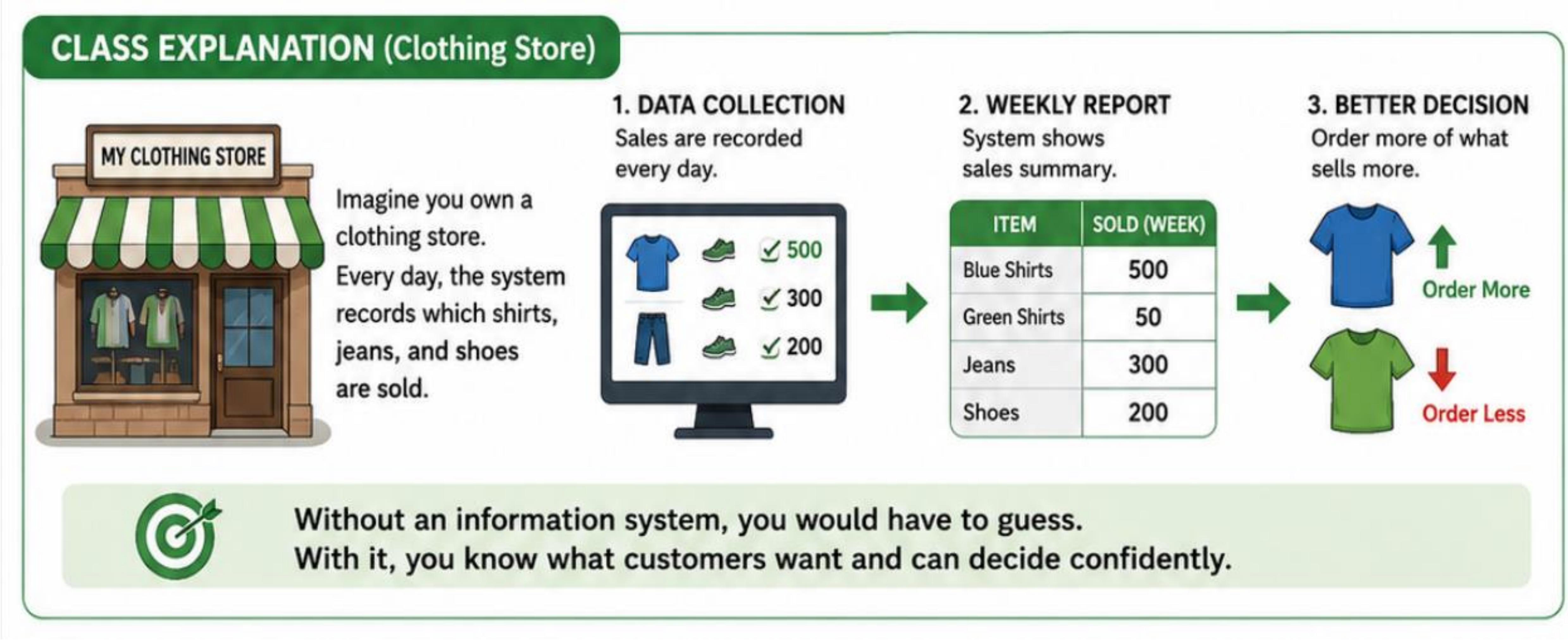
Amazon sends more umbrella stock to warehouses near that city.

4. BETTER RESULT

Customers get what they need on time. Sales and customer satisfaction increase.

Information Systems help Amazon (and other companies) make the right decisions at the right time – improving sales, reducing costs, and keeping customers happy.

2. Increases efficiency through automation



REAL-WORLD EXAMPLE – PAYTM / UPI PAYMENT

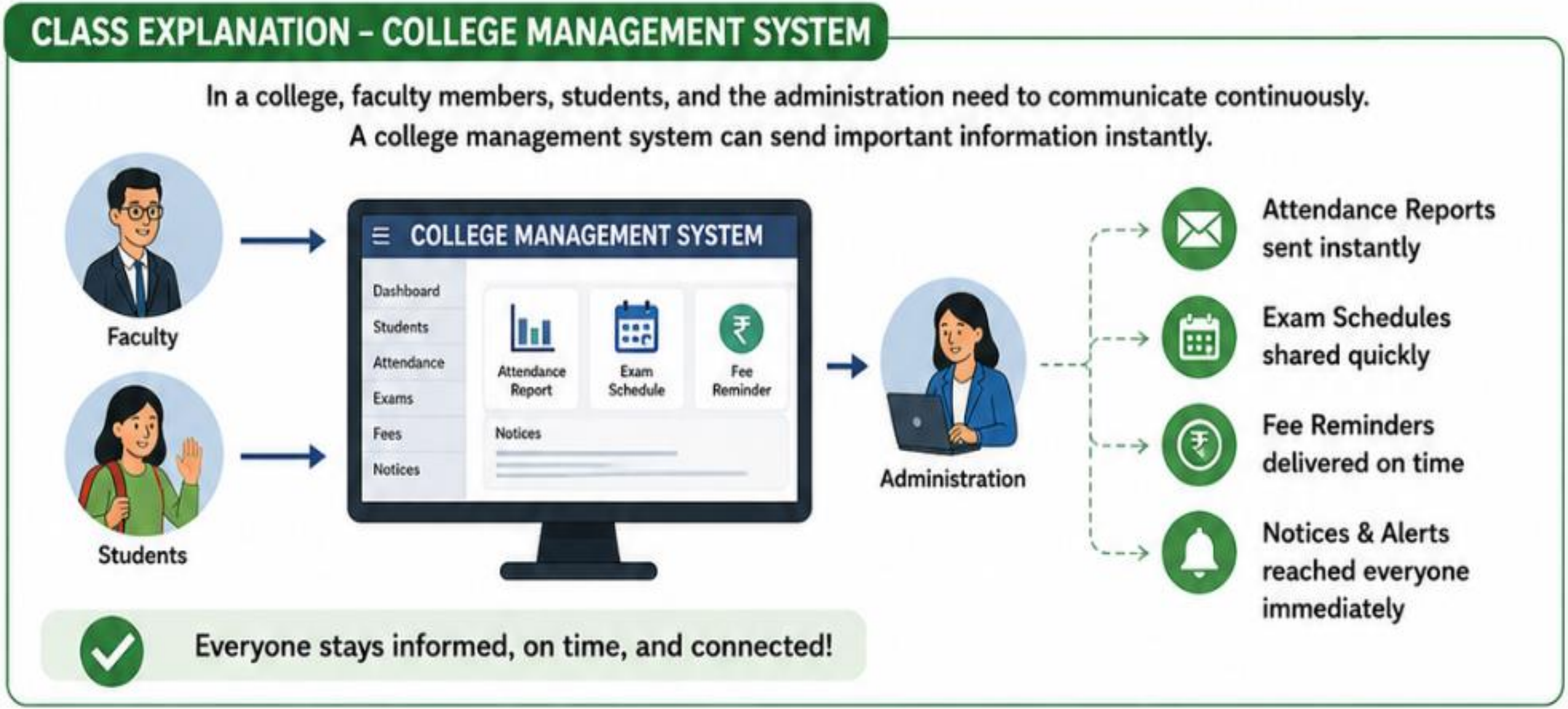
When you pay ₹500 through Paytm or any UPI app, the system automatically does everything in just a few seconds.

1. Verify Account
2. Transfer Money
3. Update Both Accounts
4. Send Confirmation

All this happens in just a few seconds!

No bank employee manually processes each payment.

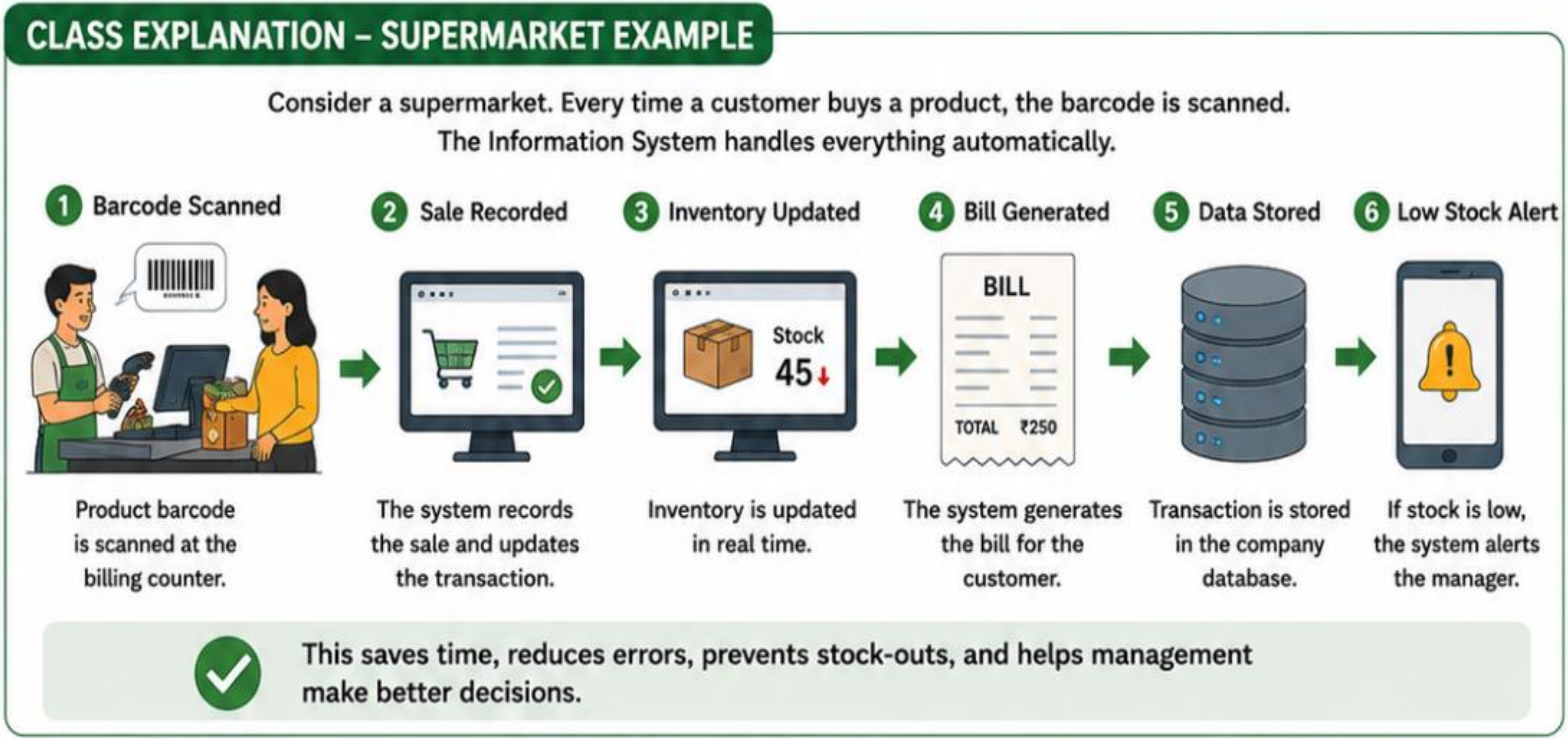
3. Enhances Communication



REAL-WORLD EXAMPLE – GOOGLE WORKSPACE

A company with employees in Delhi, Mumbai, and Bengaluru can communicate and collaborate easily using Google Workspace tools.

4. Supports Daily Business Operations



REAL-WORLD EXAMPLE – HOSPITAL INFORMATION SYSTEM

When a patient visits a hospital, the Information System manages and connects all important data.

Medical History
Past illnesses, allergies, treatments, and more.

Appointments
Book, reschedule, and manage appointments.

Prescriptions
Doctors add prescriptions and medicines.

PATIENT RECORD

Billing Details
Bills are generated and payment history stored.

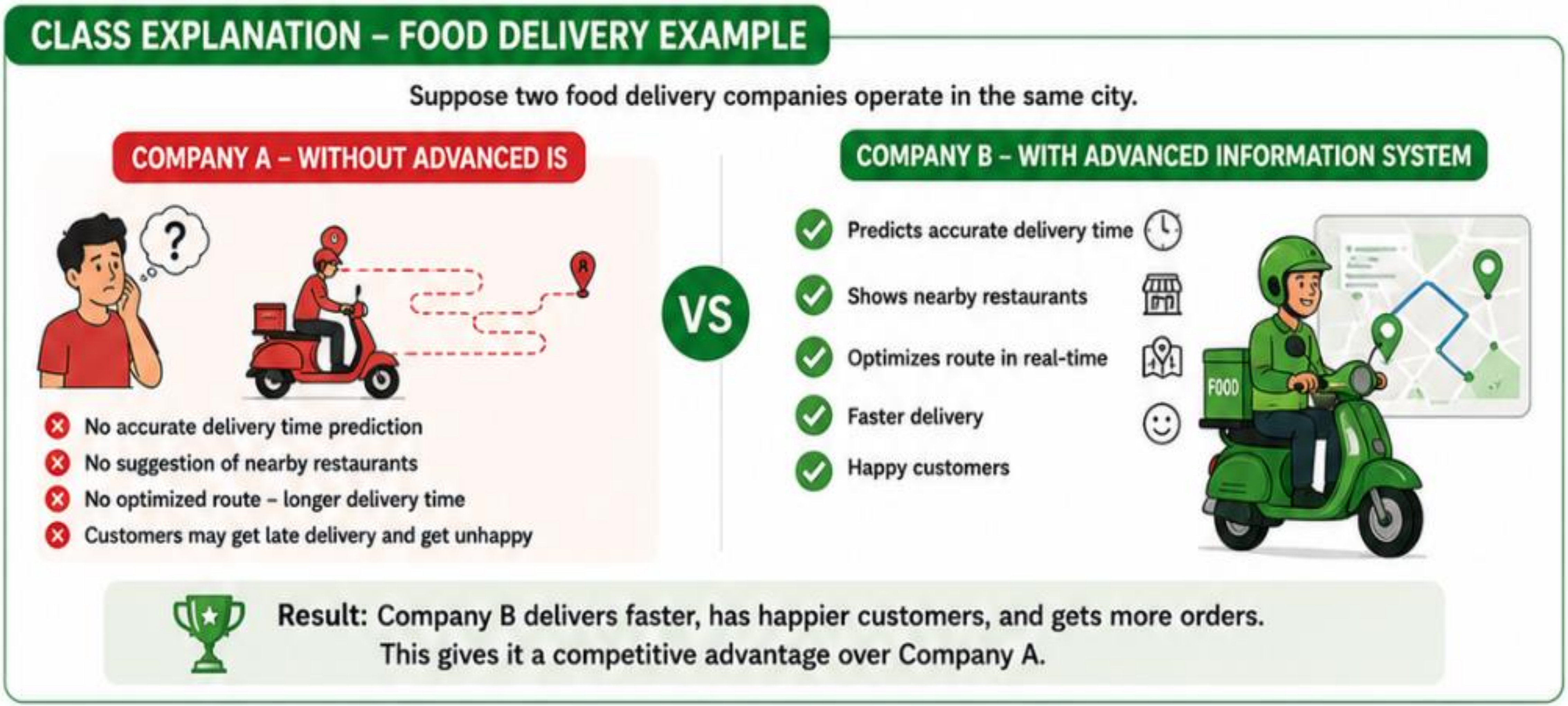
Lab Reports
Test results are stored and updated.

Insurance Information
Insurance details are saved for easy claims.

Doctors can access complete patient records instantly, anytime, anywhere.

This improves coordination, reduces paperwork, avoids errors, and improves quality of patient care.”

5. Provides Competitive Advantage



REAL-WORLD EXAMPLES

1 GOOGLE

Smart Search. Better Results.

information systems

- Understands what you mean
- Uses AI & Information Systems to find the most relevant results
- Delivers accurate results in fractions of a second

Users get the right information quickly, so Google stays the world's most popular search engine.

2 AMAZON RECOMMENDATIONS

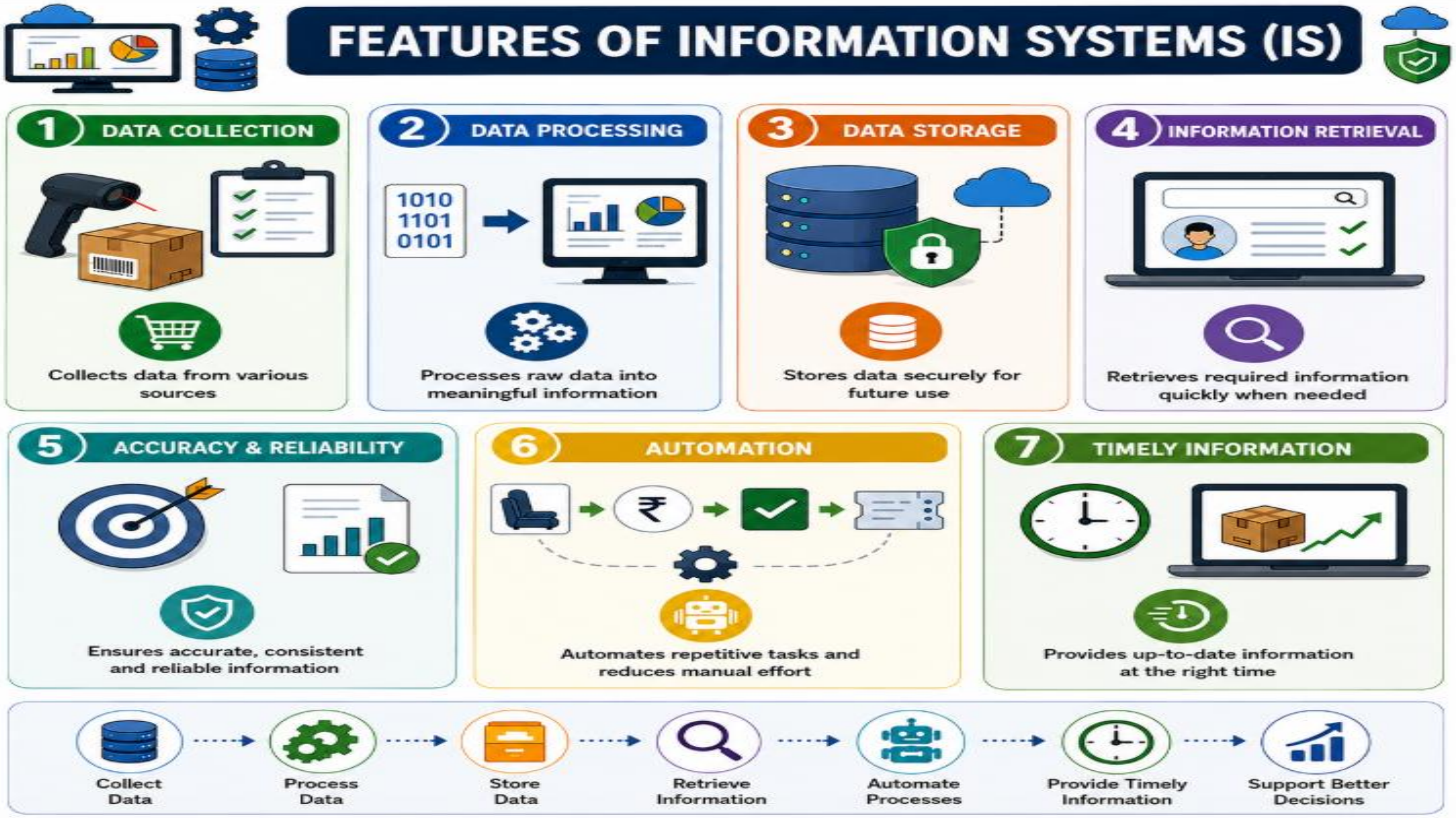
Personalized. Relevant. Engaging.

amazon

Customers who bought this item also bought...

- Collects & analyzes customer behavior (searches, views, purchases)
- Information System finds patterns & predicts what you might like
- Shows personalized recommendations to increase sales and customer satisfaction

More engagement, more sales, and loyal customers – a strong competitive advantage for Amazon.


FEATURES OF INFORMATION SYSTEMS (IS)
1 DATA COLLECTION
Collects data from various sources
2 DATA PROCESSING
1010
1101
0101
Processes raw data into meaningful information
3 DATA STORAGE
Stores data securely for future use
4 INFORMATION RETRIEVAL
Retrieves required information quickly when needed
5 ACCURACY & RELIABILITY
Ensures accurate, consistent and reliable information
6 AUTOMATION
₹
Automates repetitive tasks and reduces manual effort
7 TIMELY INFORMATION
Provides up-to-date information at the right time
Collect Data
Process Data
Store Data
Retrieve Information
Automate Processes
Provide Timely Information
Support Better Decisions

Which sequence correctly represents the logical flow of information within most Information Systems?

A. Retrieval → Collection → Storage → Processing

B. Collection → Processing → Storage → Retrieval

C. Processing → Collection → Retrieval → Storage

D. Storage → Collection → Processing → Retrieval

A university system automatically generates attendance reports from daily biometric scans. Which combination of features is involved?

A. Data Collection and Data Processing

B. Data Storage and Information Retrieval

C. Automation and Timely Information only

D. Accuracy & Reliability only

Which feature enables organizations to respond rapidly to changing market conditions?

A. Data Storage

B. Timely Information

C. Information Retrieval

D. Data Collection

Which statement best distinguishes Data Processing from Information Retrieval?

A. Processing stores data permanently, while retrieval deletes data.

B. Processing transforms raw data into information, while retrieval accesses stored information when needed.

C. Processing collects data from users, while retrieval validates data accuracy.

D. Processing automates tasks, while retrieval provides timely information only.